PORTFOLIO PERSPECTIVE: RISK-RETURN TRADE-OFF, DOWNSIDE PROTECTION, MODERN PORTFOLIO THEORY

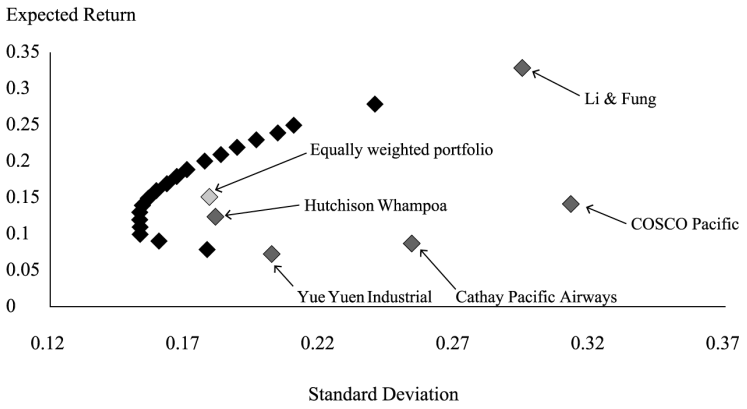
Learning Outcome

- describe the portfolio approach to investing

In the previous section we compared an equally weighted portfolio to the selection of a single security. In this section we examine additional combinations of the same set of shares and observe the trade-offs between portfolio volatility of returns and expected return (for short, their risk–return trade-offs). If we select the portfolios with the best combination of risk and return (taking historical statistics as our expectations for the future), we produce the set of portfolios shown in Exhibit 5.

Exhibit 5:

Optimal Portfolios for Sample of HKSE Listed Shares



▼ Expand / collapse Extended Description

This graph shows the optimal portfolios for five HKSE listed shares. Expected return is in the Y-axis and standard deviation is in the X-axis. An equally weighted portfolio of the five shares has an expected return of 15.1% and a standard deviation of 17.9%. However, a portfolio with 25% in Yue Yuen Industrial, 3% in Cathay Pacific, 52% in Hutchison Whampoa, 20% in Li & Fung, and 0% in COSCO Pacific produces an expected return of 15.1% and a standard deviation of 15.6%.

Source: Thomson Reuters Datastream.

In addition to illustrating that the diversified portfolio approach reduces risk, Exhibit 5 also shows that the composition of the portfolio matters. For example, an equally weighted portfolio (20 percent of the portfolio in each security) of the five shares has an expected return of 15.1 percent and a standard deviation of 17.9 percent. Alternatively, a portfolio with 25 percent in Yue Yuen Industrial (Holdings), 3 percent in Cathay Pacific, 52 percent in Hutchison Whampoa, 20 percent in Li & Fung, and 0 percent in COSCO Pacific produces a portfolio with an expected return of 15.1 percent and a standard deviation of 15.6 percent. Compared to a simple equally weighted portfolio, this provides an improved trade-off between risk and return because a lower level of risk was achieved for the same level of return.

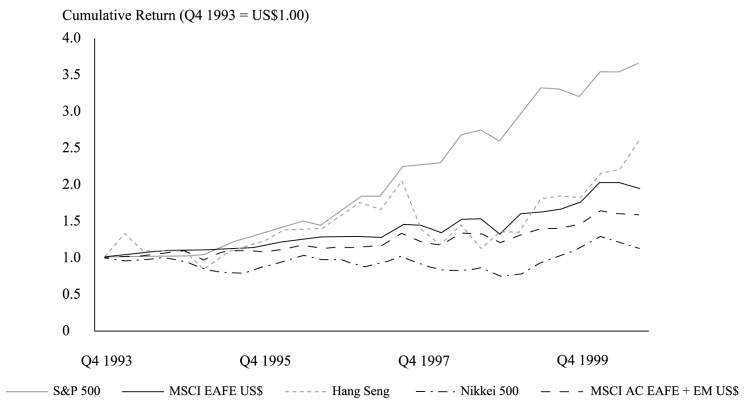
Historical Portfolio Example: Not Necessarily Downside Protection

assets will go up in value at the same time that another will go down in value. The fact that these may offset each other creates the potential diversification benefit we attribute to portfolios. However, an important issue is that the co-movement or correlation pattern of the securities' returns in the portfolio can change in a manner unfavorable to the investor. We use historical return data from a set of global indexes to show the impact of changing co-movement patterns.

When we examine the returns of a set of global equity indexes over the last 15 years, we observe a reduction in the diversification benefit due to a change in the pattern of co-movements of returns. Exhibit 6 and Exhibit 7 show the cumulative returns for a set of five global indexes[7] for two different time periods. Comparing the first time period, from Q4 1993 through Q3 2000 (as shown in Exhibit 6), with the last time period, from Q1 2006 through Q1 2009 (as shown in Exhibit 7), we show that the degree to which these global equity indexes moved together increased over time.

Exhibit 6:

Returns to Global Equity Indexes Q4 1993–Q3 2000



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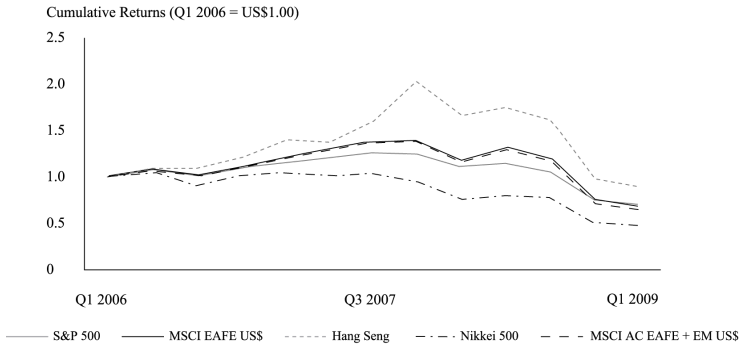
This graph shows the cumulative returns for a set of five global equity indexes for the time period from Q4 1993 through Q3 2000. The five global indexes are S&P 500, MSCI EAFE US$, Hang Seng, Nikkei 500, and MSCI AC EAFE + EM US$. Cumulative returns (Q4 1993 = US$1.00) is in the Y-axis and quarter is in the X-axis.

Source: Thomson Reuters Datastream.

Exhibit 7:

Returns to Global Equity Indexes Q1 2006–Q1 2009



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This graph shows the cumulative returns for a set of five global equity indexes for the time period from Q1 2006 through Q1 2009. The five global indexes are S&P 500, MSCI EAFE US$, Hang Seng, Nikkei 500, and MSCI AC EAFE + EM US$. Cumulative returns (Q1 2006 = US$1.00) is in the Y-axis and quarter is in the X-axis. The latter part of this time period, from Q4 2007 to Q1 2009, was a period of dramatic declines in global share prices.

The latter part of the second time period, from Q4 2007 to Q1 2009, was a period of dramatic declines in global share prices. Exhibit 8 shows the mean annual returns and standard deviation of returns for this time period.

Exhibit 8:

Returns to Global Equity Indexes

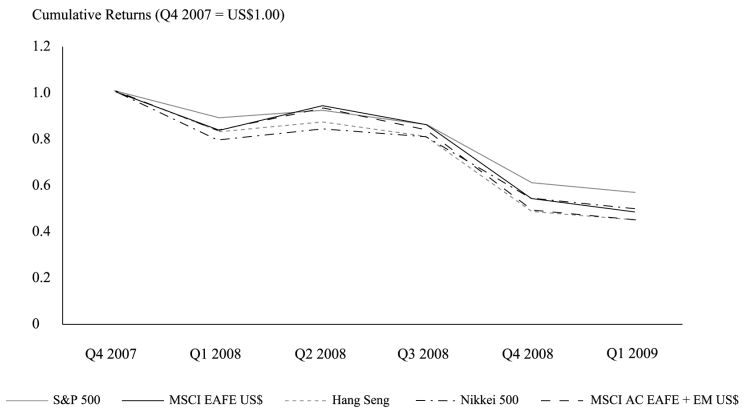
	Q4 1993–Q3 2000		Q1 2006–Q1 2009		Q4 2007–Q1 2009	
Global Index	**Mean**	**Stand. Dev.**	**Mean**	**Stand. Dev.**	**Mean**	**Stand. Dev.**
S&P 500	20.5%	13.9%	–6.3%	21.1%	–40.6%	23.6%
MSCI EAFE US$	10.9	14.2	–3.5	29.4	–48.0	35.9
Hang Seng	20.4	35.0	5.1	34.2	–53.8	34.0
Nikkei 500	3.3	18.0	–13.8	27.6	–48.0	30.0
MSCI AC EAFE + EM US$	7.6	13.2	–4.9	30.9	–52.0	37.5
Randomly selected index	12.6%	18.9%	–4.7%	28.6%	–48.5%	32.2%
Equally weighted portfolio	12.6%	14.2%	–4.7%	27.4%	–48.5%	32.0%
Diversification ratio		75.1%		95.8%		99.4%

Source: Thomson Reuters Datastream.

During the period Q4 2007 through Q1 2009, the average return for the equally weighted portfolio, including dividends, was −48.5 percent. Other than reducing the risk of earning the return of the worst performing market, the diversification benefits were small. Exhibit 9 shows the cumulative quarterly returns of each of the five indexes over this time period. All of the indexes declined in unison. The lesson is that although portfolio diversification generally does reduce risk, it does not necessarily provide the same level of risk reduction during times of severe market turmoil as it does when the economy and markets are operating 'normally'. In fact, if the economy or markets fail totally (which has happened numerous times around the world), then diversification is a false promise. In the face of a worldwide contagion, diversification was ineffective, as illustrated at the end of 2008.

Exhibit 9:

Return to Global Equity Indexes Q4 2007–Q1 2009



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This graph shows the cumulative quarterly returns of each of five global equity indexes over the Q4 2007 through Q1 2009 time period. All of the indexes declined in unison during this time period. In the face of a worldwide contagion, diversification was ineffective, as illustrated at the end of 2008. Cumulative returns (Q4 2007 = US$1.00) is in the Y-axis and quarter is in the X-axis. The five global equity indexes are S&P 500, MSCI EAFE US$, Hang Seng, Nikkei 500, and MSCI AC EAFE + EM US$.

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Knowledge Check

Portfolios are *most likely* to provide:

A. risk reduction.
B. risk elimination.
C. downside protection.

Hide Solution

Solution:

A is correct. Combining assets into a portfolio should reduce the portfolio's volatility. However, the portfolio approach does not necessarily provide downside protection or eliminate all risk.

Portfolios: Modern Portfolio Theory

The concept of diversification has been around for a long time and has a great deal of intuitive appeal. However, the actual theory underlying this basic concept and its application to investments only emerged in 1952 with the publication of Harry Markowitz's classic article on portfolio selection.[8] The article provided the foundation for what is now known as **modern portfolio theory** (MPT). The main conclusion of MPT is that investors should not only hold portfolios but should also focus on how individual securities in the portfolios are related to one another. In addition to the diversification benefits of portfolios to investors, the work of William **Sharpe (1964)**, John **Lintner (1965)**, and Jack **Treynor (1961)** demonstrated the role that portfolios play in determining the appropriate individual asset risk premium (i.e., the return in excess of the risk-free return expected by investors as compensation for the asset's risk). According to capital market theory, the priced risk of an individual security is affected by holding it in a well-diversified portfolio. The early research provided the insight that an asset's risk should be measured in relation to the remaining systematic or non-diversifiable risk, which should be the only risk that affects the asset's price. This view of risk is the basis of the capital asset pricing model, or CAPM, which is discussed in greater detail in other readings. Although MPT has limitations, the concepts and intuitions illustrated in the theory continue to be the foundation of knowledge for portfolio managers.

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